INTERFERENCE

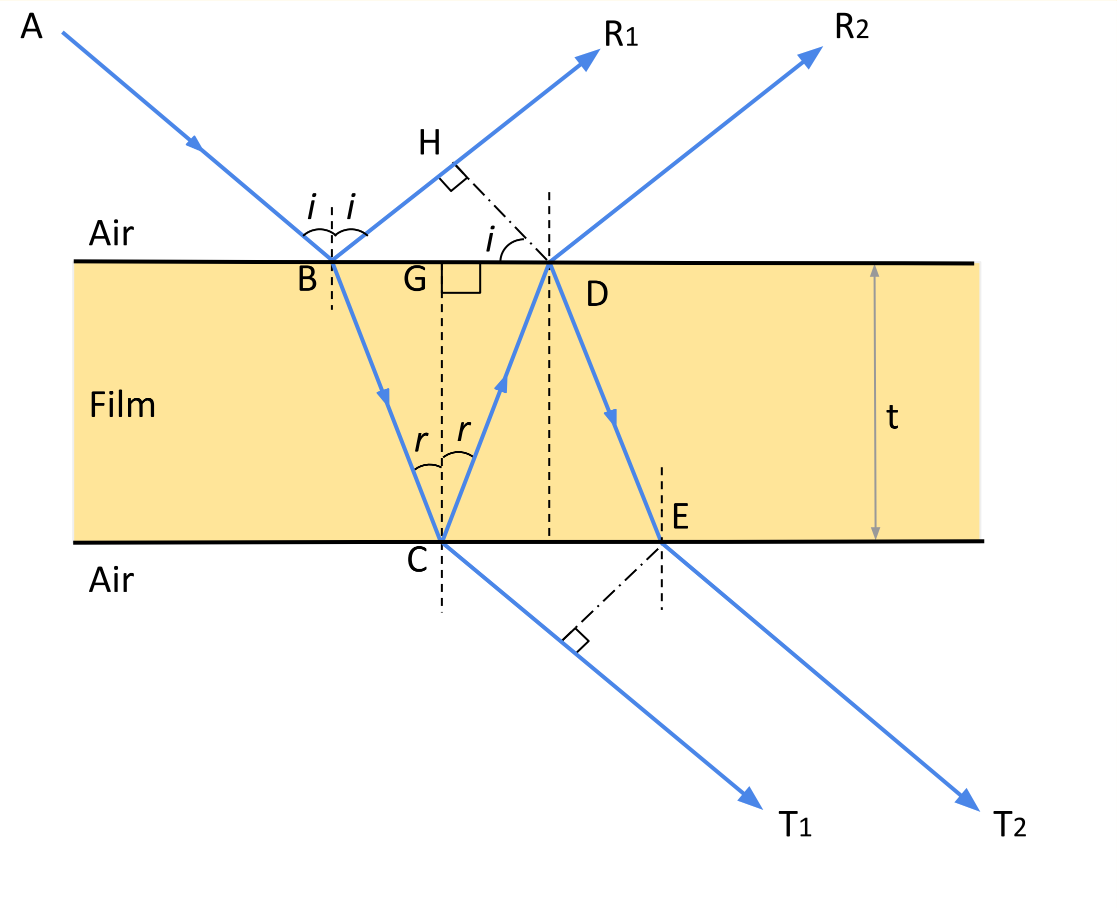
Interference in a thin, transparent film.

7 times for 4-8 Marks

- Refraction while going from a ***rarer to denser*** medium generates an additional ***phase difference of π***, which results in additional ***path difference of λ/2.***
- For a ***reflected system***:

1. Condition for ***maxima*: $2\mu t(\cos r) = (2n-1)(\lambda/2)$**
2. Condition for ***minima***: **$2\mu t(\cos r) = n\lambda$**

- For a transmitted system:
  1. Condition for ***maxima***: **$2\mu t(\cos r) = n\lambda$**
  2. Condition for ***minima***: **$2\mu t(\cos r) = (2n-1)(\lambda/2)$**
- A ***reflected*** system has ***higher contrast*** in the ***intensities of the maxima and minima***, while a ***transmitted*** system has ***poor contrast***.



Wedge-shaped film.

5 times for 3-7 Marks

- Condition for constructive interference(***maxima***):

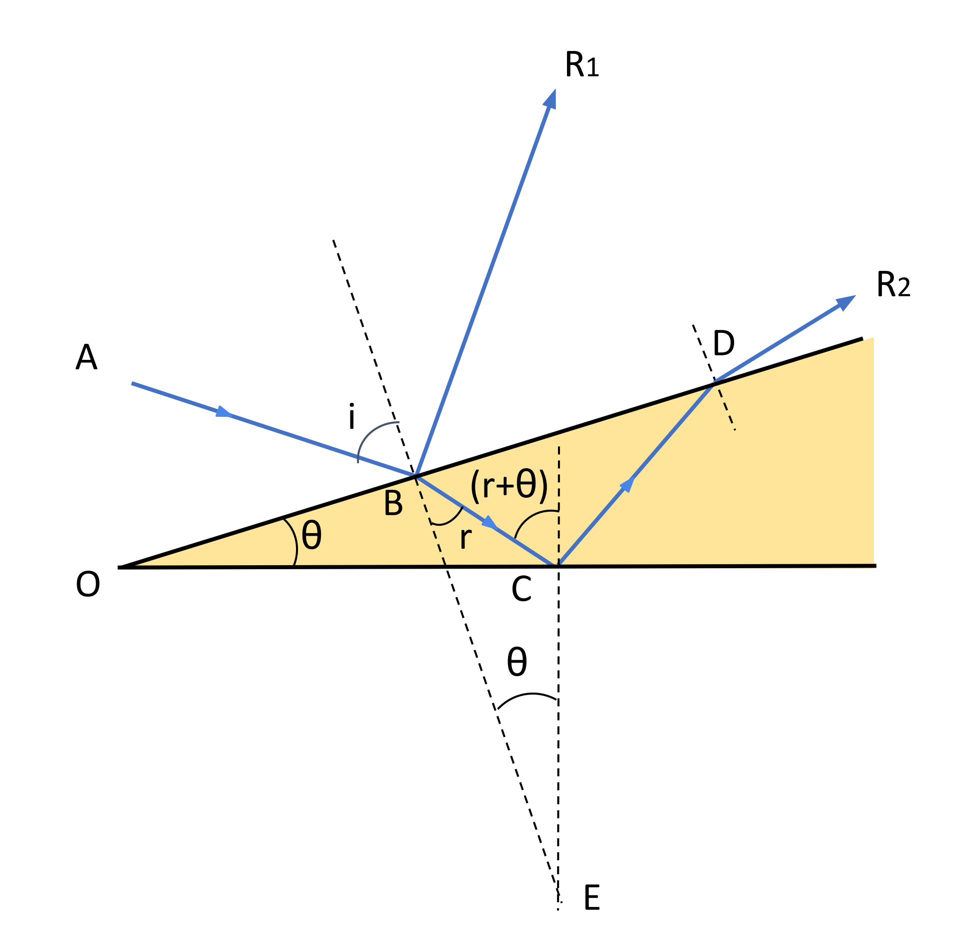
  $\mathbf{2\mu t(\cos(r+\theta)) = (2n-1)(\lambda/2)}$

  where n = 1,2,3,4...

- Condition for destructive interference(***minima***):

  $\mathbf{2\mu t(\cos(r+\theta)) = n\lambda}$

  Where n = 0,1,2,3...



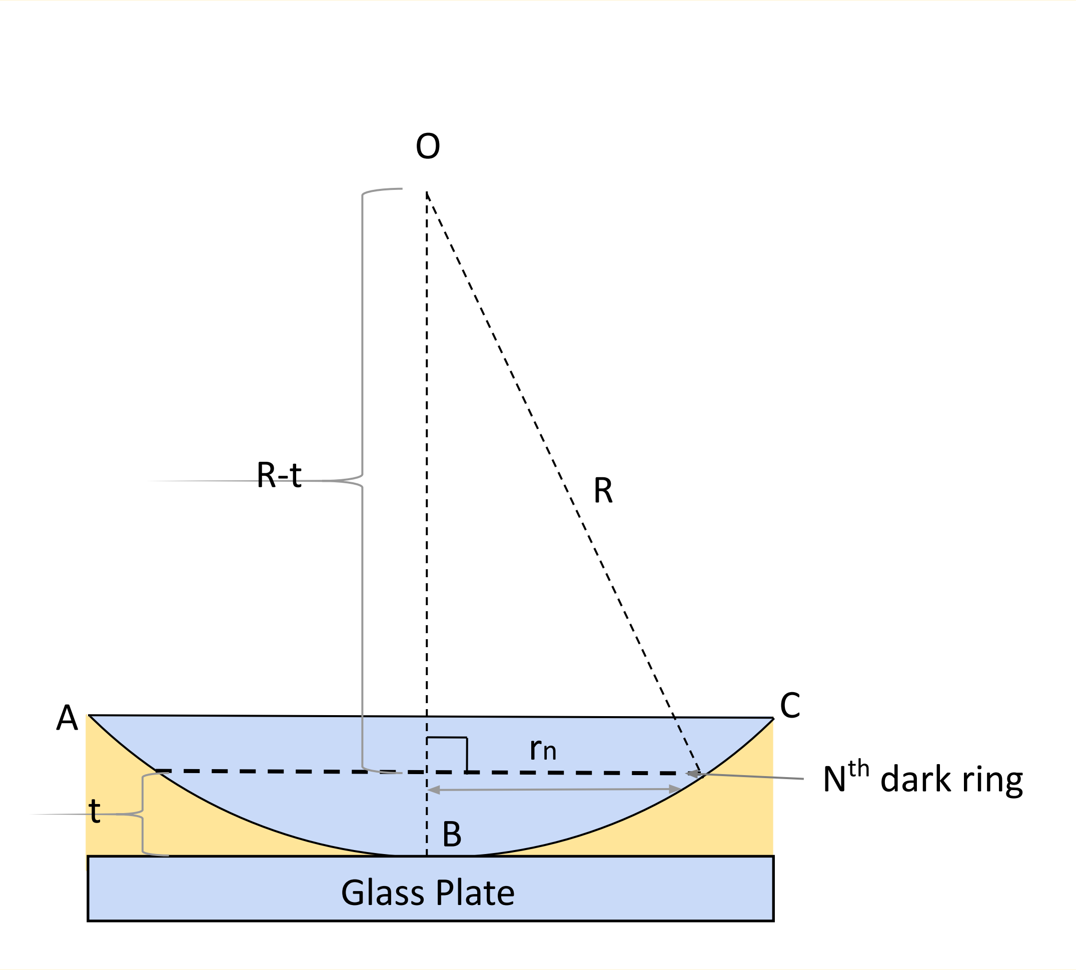
Newton's Rings

5 times for 3-7 Marks

- Newton’s Rings (NR) setup consist of the plano-convex lens placed on a glass plate.

- Once the setup is illuminated alternate bright and dark rings are formed due to amplitude division type of interference in the thin film that is created between the plano-convex lens and glass plate.

- For a dark ring: $\Delta = (2n+1)(\lambda/2)$
- For a bright ring:
  - $\Delta = n\lambda$
  - $r_n = ((2n-1)(\lambda/2)R)^{½}$
  - Thus $r_n \alpha (2n-1)^{½}$



Circular Fringes in Newton's Rings/ Straight Fringes in Wedge-shaped films.

4 times for 3 Marks

- Fringes are the ***locus*** of points of e***qual thickness***.
- In Newton's Rings, the points of equal thickness lie on a ***circular path (in concentric circles)***.
- In wedge-shaped films, points of equal thickness are ***straight, parallel lines***.
- Thus, fringes in Newton's Rings are circular and concentric, while fringes in wedge-shaped films are straight and parallel.

Why is the centre always dark, in Newton's Rings?

1 time for 3 Marks

- Condition for bright rings: $2\mu t(\cos(r+\theta)) = (2n+1)(\lambda/2)$
- Condition for dark rings: $2\mu t(\cos(r+\theta)) = n\lambda$
- Thickness at point of contact $(t) = 0$
- For large lens radius, $\theta = 0$, and for normal incidence, $r = 0$
- ***New*** condition for bright rings: $2\mu t = (2n+1)(\lambda/2)$
- ***New*** condition for dark rings: $2\mu t = n\lambda$
- t = 0 only ***satisfies*** the new ***condition for dark rings***, hence the centre is always dark.

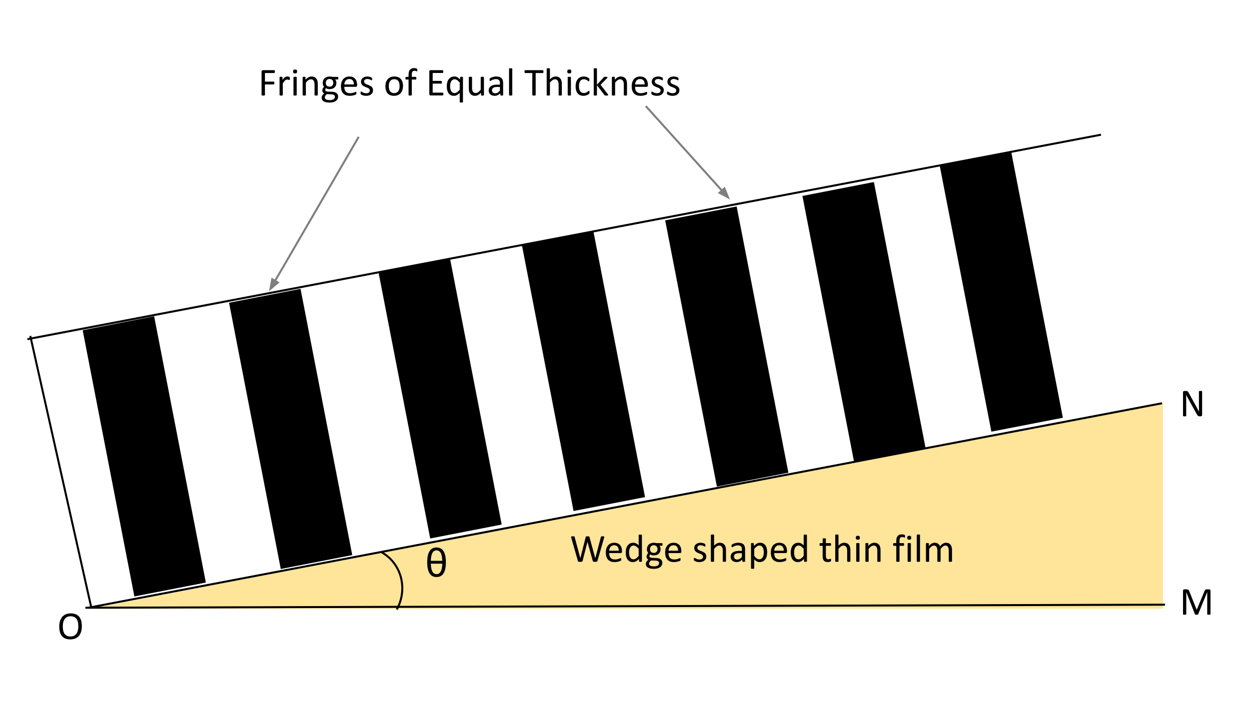
Liquid introduced in Newton's Rings

- The ***center*** of the Newton's Rings experiment, when perfomed under ***normal conditions (air)***, is always ***dark***.
- If a liquid, whose ***μ*** is ***greater than the lens*** but ***smaller than plate***, is used, the ***centre*** becomes ***bright.***
- This is because light is going from a r***arer to denser medium twice***, which results in a ***phase change of π*** twice.
- This collectively results in ***no phase change*** at all.
- Hence, constructive interference occurs which gives rise to a bright centre spot.

Applications of Diffraction
Testing Optical Flatness

1 time for 3 Marks

- Based on ***Interference.***
- If ON and OM are flat, they form a ***wedge-shaped film interference*** system, which varies consistently in thickness.
- ***Fringe pattern*** is tested.
- If the fringes have ***equal thickness***, then the test surface is optically flat.



Applications of Diffraction

Origin of colour in thin films

4 times for 3 Marks

- When white light is incident on a film, ***different colours travel different lengths of path*** in the film.
- Thus, ***different colours are reflected from different points*** on the film.
- Some ***colour interferes constructively*** and is ***intensified***, while the ***rest fade out***.
- This is how colour originates in thin films.

Applications of Diffraction

Anti-Reflecting Films

4 times for 3-8 Marks

- Light is ***lost due to reflection*** at glass-air interfaces.
- ***Hinders*** working of ***complicated optical*** devices.
- An anti-reflecting film ***prevents*** this.
- The refractive index of the film is ***greater than that of air*** but ***less than that of the material*** it is coated on.